

DBMS ASSIGNMENT – II

Entity Relationship (ER) Diagram and Relational Schema

Application: Online Shopping System

Objective

To design a complete Entity Relationship (ER) Diagram for an Online Shopping System, identify entities, attributes, primary keys, relationships and cardinalities, and convert the ER model into a relational schema with appropriate primary and foreign keys.

1. Entities and Attributes

The Online Shopping System contains the following entities. Each entity has a primary key (PK), and foreign keys (FK) are used to represent relationships between related tables.

Entity	Primary Key	Attributes
CUSTOMERS	customer_ID	customer_ID, first_name, last_name, email, phone_number, password_hash, registration_date
ADDRESSES	address_ID	address_ID, address_line, city, state, pincode, address_type, customer_ID (FK)
SHOPPING_CART	cart_ID	cart_ID, created_date, customer_ID (FK)
CART_ITEMS	cart_item_ID	cart_item_ID, quantity, cart_ID (FK), product_ID (FK)
ORDERS	order_ID	order_ID, order_date, status, total_amount, customer_ID (FK)
ORDER_ITEMS	order_item_ID	order_item_ID, quantity, unit_price, order_ID (FK), product_ID (FK)
PRODUCTS	product_ID	product_ID, product_name, brand, price, stock_quantity, description, category_ID (FK)
CATEGORIES	category_ID	category_ID, category_name, description
REVIEWS	review_ID	review_ID, rating, comment, review_date, customer_ID (FK), product_ID (FK)
PAYMENTS	payment_ID	payment_ID, transaction_ID, payment_date, amount, status, order_ID (FK)
DELIVERIES	delivery_ID	delivery_ID, tracking_number, delivery_status, shipped_date, address_ID (FK), order_ID (FK)

2. Relationships and Cardinalities

Relationship	Cardinality	Explanation
Customer — Address	1:M	One customer can have multiple addresses; each address belongs to one customer.
Customer — Shopping Cart	1:1	One customer has one active shopping cart.
Shopping Cart — Cart Items	1:M	One cart can contain multiple cart items.
Product — Cart Items	1:M	One product can appear in multiple cart items.
Customer — Orders	1:M	One customer can place multiple orders.
Order — Order Items	1:M	One order contains multiple order items.
Product — Order Items	1:M	One product can appear in multiple order items.
Category — Product	1:M	One category can contain multiple products; each product belongs to one category.
Customer — Reviews	1:M	One customer can write multiple reviews.
Product — Reviews	1:M	One product can receive multiple reviews.

Order — Payment	1:1	One order is associated with one payment.
Order — Delivery	1:1	One order is associated with one delivery record.
Address — Delivery	1:M	One address can be used for multiple deliveries.

3. ER Diagram

The ER Diagram below represents the entities, attributes, relationships and cardinalities of the Online Shopping System.

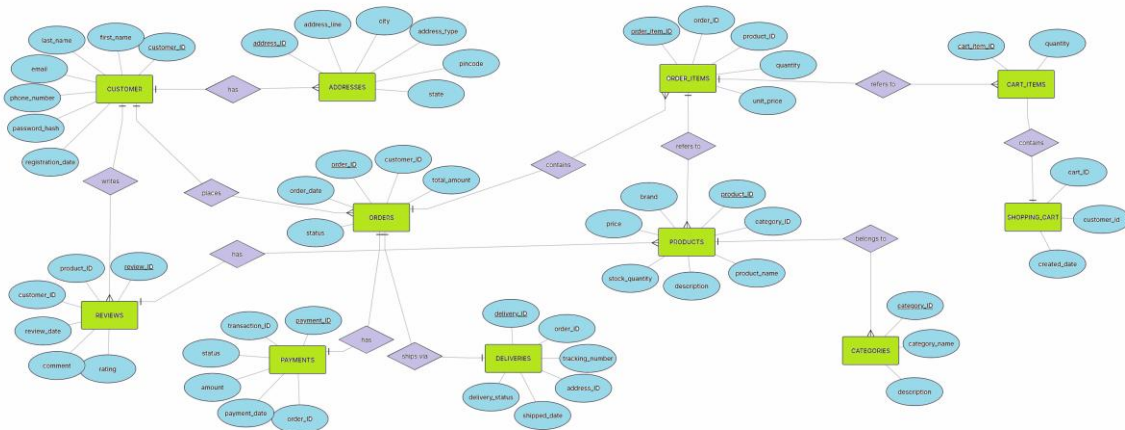


Figure 1: Entity Relationship Diagram – Online Shopping System

4. Relational Schema

The ER model is converted into the following relational schema. Primary keys are marked as PK and foreign keys as FK.

CUSTOMERS (customer_ID (PK), first_name, last_name, email, phone_number, password_hash, registration_date)

ADDRESSES (address_ID (PK), address_line, city, state, pincode, address_type, customer_ID (FK))

SHOPPING_CART (cart_ID (PK), created_date, customer_ID (FK))

CART_ITEMS (cart_item_ID (PK), quantity, cart_ID (FK), product_ID (FK))

ORDERS (order_ID (PK), order_date, status, total_amount, customer_ID (FK))

ORDER_ITEMS (order_item_ID (PK), quantity, unit_price, order_ID (FK), product_ID (FK))

PRODUCTS (product_ID (PK), product_name, brand, price, stock_quantity, description, category_ID (FK))

CATEGORIES (category_ID (PK), category_name, description)

REVIEWS (review_ID (PK), rating, comment, review_date, customer_ID (FK), product_ID (FK))

PAYMENTS (payment_ID (PK), transaction_ID, payment_date, amount, status, order_ID (FK))

DELIVERIES (delivery_ID (PK), tracking_number, delivery_status, shipped_date, address_ID (FK), order_ID (FK))

5. Primary Key and Foreign Key Identification

Table	Primary Key (PK)	Foreign Key(s) (FK)
CUSTOMERS	customer_ID	—
ADDRESSES	address_ID	customer_ID → CUSTOMERS(customer_ID)
SHOPPING_CART	cart_ID	customer_ID → CUSTOMERS(customer_ID)
CART_ITEMS	cart_item_ID	cart_ID → SHOPPING_CART(cart_ID); product_ID → PRODUCTS(product_ID)
ORDERS	order_ID	customer_ID → CUSTOMERS(customer_ID)
ORDER_ITEMS	order_item_ID	order_ID → ORDERS(order_ID); product_ID → PRODUCTS(product_ID)
PRODUCTS	product_ID	category_ID → CATEGORIES(category_ID)
CATEGORIES	category_ID	—
REVIEWS	review_ID	customer_ID → CUSTOMERS(customer_ID); product_ID → PRODUCTS(product_ID)
PAYMENTS	payment_ID	order_ID → ORDERS(order_ID)
DELIVERIES	delivery_ID	address_ID → ADDRESSES(address_ID); order_ID → ORDERS(order_ID)

6. Relationship Explanation

Customer and Address: The relationship is 1:M because a customer may save multiple addresses, such as home and office, while each address is associated with one customer.

Customer and Shopping Cart: The relationship is 1:1 because each customer has one active shopping cart in this design.

Shopping Cart and Cart Items: The relationship is 1:M because a shopping cart can contain multiple cart items.

Cart Items and Products: The relationship is M:1 from Cart Items to Products. CART_ITEMS acts as an associative entity between shopping carts and products.

Customer and Orders: The relationship is 1:M because a customer can place many orders over time.

Orders and Order Items: The relationship is 1:M because one order can contain multiple products, represented through ORDER_ITEMS.

Order Items and Products: The relationship is M:1 from ORDER_ITEMS to PRODUCTS. ORDER_ITEMS resolves the many-to-many relationship between orders and products.

Categories and Products: The relationship is 1:M because one category can contain many products, while each product belongs to one category. Therefore, category_ID is an FK in PRODUCTS.

Customer and Reviews: The relationship is 1:M because a customer may write multiple reviews.

Product and Reviews: The relationship is 1:M because a product can receive reviews from multiple customers.

Orders and Payments: The relationship is 1:1 in this simplified model because each order has one payment record.

Orders and Deliveries: The relationship is 1:1 in this simplified model because each order has one delivery record.

Addresses and Deliveries: The relationship is 1:M because the same saved address may be used for deliveries for multiple orders.

7. M:N Relationships and Associative Tables

Two important many-to-many relationships in the Online Shopping System are resolved using associative entities:

Entities	Relationship	Associative Table	Purpose
ORDERS ↔ PRODUCTS	M:N	ORDER_ITEMS	Stores quantity and unit_price for each product in an order.
SHOPPING_CART ↔ PRODUCTS	M:N	CART_ITEMS	Stores the quantity of each product currently present in a cart.

Reference Relational Schema Diagram

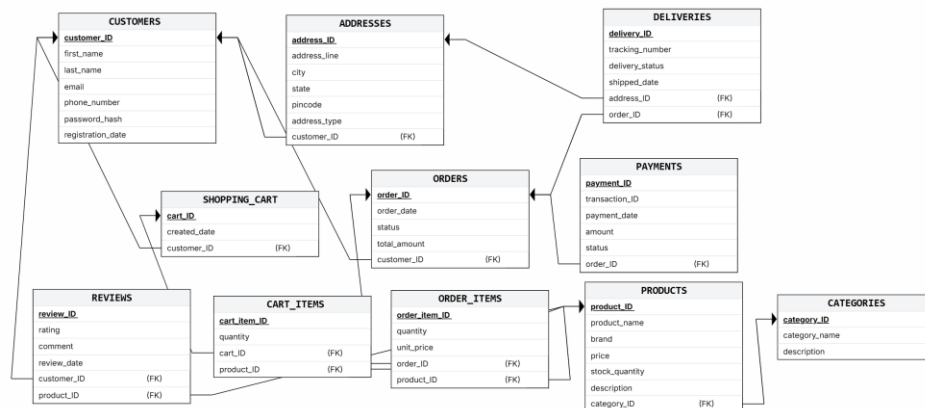


Figure 2: Relational Schema Reference